

AI Stock Market Prediction & Analysis Report

Symbol: AAPL | Generated on: 2026-07-01 21:03:43

Stock Market Overview

As of the latest data point (2017-11-10), **AAPL** closed at **\$174.67**, representing a change of **-0.58 (-0.33%)** from the previous trading session. The stock's technical setup is detailed below, showing key momentum indices and volatility metrics.

Indicator Name	Value	Interpretation
Close Price	\$174.67	Latest session close price
RSI (14)	85.83	Overbought (>70)
MACD	4.9793	Bullish Crossover
20-Day SMA	\$164.52	Short-term price trend support
50-Day SMA	\$159.59	Medium-term price trend support
Annual Volatility	21.47%	Annualized standard deviation of returns

Machine Learning Next-Day Predictions

Our models predict the next trading day's close price using sequential modeling. Confidence scores represent aggregated R^2 accuracy and mean absolute percentage error (MAPE) metrics.

Model Name	Next Day Prediction	Confidence Score
Linear Regression	\$2.14	3.4%
Random Forest	\$45.28	33.2%
XGBoost	\$43.50	32.3%
LSTM Deep Learning	\$45.69	20.3%

Model Performance Evaluation Cache

Evaluation metrics are calculated chronologically on the 20% test partition. A higher R^2 indicates better trend correlation, while lower MAE/RMSE indicate lower absolute price errors.

Model Name	MAE (\$)	RMSE (\$)	R^2 Score	MAPE (%)
Random Forest	46.5865	55.8079	-2.2799	44.71%
XGBoost	47.9659	57.1604	-2.4408	46.24%
Linear Regression	87.6948	93.5221	-8.2108	94.29%

LSTM	99.7023	100.6761	-115.8477	66.17%
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Disclaimer

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